THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE FULL SYLLABUS PAPER SET 02

Full Syllabus (Chapters 1–11) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. Why should we never taste unknown chemicals? 1

(a) They may be harmful **(b)** They are always sour
(c) They have no taste **(d)** They are costly

2. The word ‘acid’ comes from the Latin word ‘acere’ which means: 1

(a) sour **(b)** salty **(c)** bitter **(d)** sweet

3. Two cells joined end to end form a: 1

(a) switch **(b)** battery **(c)** fuse **(d)** diode

4. The metal used in wires for good conduction is mostly: 1

(a) glass **(b)** rubber **(c)** copper **(d)** wood

5. The gas essential for respiration, a non-metal, is: 1

(a) helium **(b)** hydrogen **(c)** oxygen **(d)** neon

6. Which is used in cooking vessels because it conducts heat well? 1

(a) Wood **(b)** Aluminium **(c)** Plastic **(d)** Glass

7. Which is a sudden, fast change? 1

(a) Weathering **(b)** Bursting of a firecracker
(c) Formation of soil **(d)** Rusting

8. The change of marble into lime when heated strongly is: 1

(a) reversible only **(b)** dissolving **(c)** chemical **(d)** physical

9. The best way to resist wrong peer pressure is to: 1

(a) stop studying **(b)** run away
(c) think independently and say no **(d)** follow everyone

10. Personal hygiene during adolescence includes: 1

(a) skipping brushing **(b)** sharing towels
(c) avoiding water **(d)** regular bathing and clean clothes

11. When heated, the portion of a fluid that rises is: 1

(a) less dense **(b)** denser **(c)** colder **(d)** frozen

12. Dark-coloured objects absorb heat from the Sun: 1

(a) equally **(b)** not at all
(c) more than light-coloured objects **(d)** less than light-coloured

13. A cyclist travels 90 km at 30 km/h. The time taken is: 1

(a) 2700 hours **(b)** 3 hours **(c)** 6 hours **(d)** 4 hours

14. 54 km/h is equal to: 1

(a) 194.4 m/s **(b)** 0.9 m/s **(c)** 54 m/s **(d)** 15 m/s

15. Absorption of digested food into blood takes place mainly in the: 1

(a) stomach **(b)** small intestine **(c)** oesophagus **(d)** rectum

16. The pathway of food from mouth to anus is the: 1

(a) nervous system **(b)** respiratory tract
(c) alimentary canal **(d)** circulatory system

17. In a starch test, parts of a variegated leaf that were white show: 1

(a) no blue-black colour **(b)** red colour
(c) green colour **(d)** blue-black colour

18. Plants take in carbon dioxide mainly through tiny pores called: 1

(a) xylem **(b)** root hairs **(c)** veins **(d)** stomata

19. A shiny polished surface produces: 1

(a) regular reflection **(b)** absorbed light
(c) no reflection **(d)** only shadows

20. The Moon's phases are caused by: 1

(a) its position relative to the Sun and Earth
(b) its rotation only
(c) clouds
(d) Earth's shadow always

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 15–25 words each.

21. What is an olfactory indicator? Give an example. 2

22. What is meant by a closed circuit? 2

23. What happens when magnesium burns in air? 2

24. Why does the timing of changes vary among adolescents? 2

25. Name three processes in the water cycle. 2

26. Why is it better to use a pendulum or quartz clock than a sundial? 2

27. How do fish breathe? 2

28. What is the role of Rhizobium in legumes? 2

SECTION — C (Short Answer Type) 8 × 3 = 24

Answer in about 40–60 words each.

29. List the steps to light a bulb using a cell and two wires without a holder. 3

30. Identify the changes in photosynthesis and digestion. 3

31. Explain why water at the bottom of a pot heats first but the entire water warms. 3

32. State two traditional breathing practices and their benefits. 3

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33. A student sees the word 'STOP' in a mirror. How does it look? Why? 3

34. A traveller goes from India to Australia. Why will seasons be opposite? 3

35. Why do ants' bites burn but bee stings can be eased with mild bases? 3

OR

State two uses of bases in daily life. 3

36. Why do green plants not need to eat? 3

OR

State three differences between day and night activities in a plant. 3

SECTION — D (Long Answer Type) 4 × 5 = 20

Answer in about 80–120 words each.

37. (i) Explain how oxides of metals and non-metals differ. 5
(ii) How do you test them?
(iii) Give one example of each.

OR

(i) How do we prevent corrosion? 5
(ii) Explain three methods.
(iii) Why is prevention important?

38. (i) Explain the importance of iron and calcium. 5
(ii) Give sources.
(iii) Disorders from their deficiency.

OR

Case study: Meera feels irritable before exams and is anxious about her appearance. 5
(i) Is it normal?
(ii) Name the cause.
(iii) Suggest two remedies.
(iv) When to seek help?

39. (i) Explain simple pendulum. 5
(ii) What factors affect its time period?
(iii) How is it used?

OR

Case study: Two cars start together. Car A moves at 72 km/h; Car B at 15 m/s. 5
(i) Whose speed is greater?
(ii) Distance covered in 1 h by each.
(iii) Who is ahead after 1 h?
(iv) By how much?

40. **Case study:** Light from the Sun forms a round spot below a tree through gaps between leaves. 5
(i) Why round?
(ii) Which device works similarly?
(iii) Would size change with gap?
(iv) What does this show about light?

OR

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Case study: Torch light directed at a smooth mirror and a rough wall. 5

(i) Which gives clear image?

(ii) Why?

(iii) Name the reflections.

(iv) Give an everyday example.

END OF QUESTION PAPER **Total: 80 Marks**

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